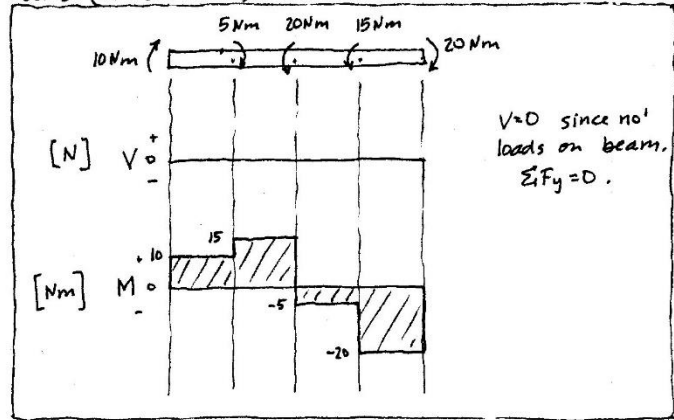
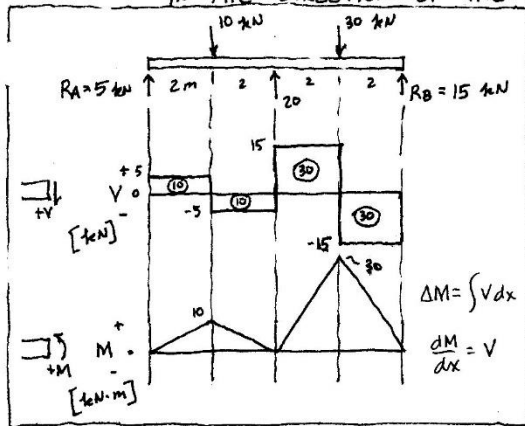


Statics Internal Forces: Intro to Shear and Bending Moment Diagrams (Review of Three Rules plus Two Simple Examples)

THE THREE BASIC RULES OF V & M DIAGRAM

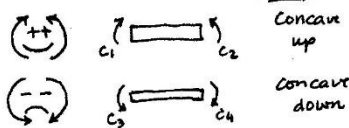
- ① POINT LOADS AND SUPPORTS PRODUCE JUMPS IN THE SHEAR DIAGRAM
IN THE DIRECTION OF THE POINT LOAD (OR SUPPORT)



- ② COUPLES CORRESPOND TO JUMPS IN THE MOMENT DIAGRAM.

Two cases, or two sign conventions, to remember:

- (a) COUPLES ACTING AT ENDS OF BEAM

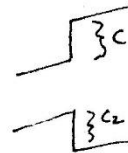


The moment curve will start or end at those non-zero values at the ends of the beam.

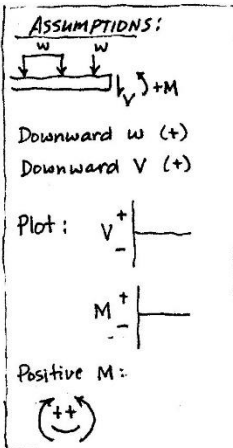
- (b) COUPLES ACTING IN INTERIOR OF BEAM

CW $\curvearrowright_{C1}$ PRODUCES (+) JUMP IN M CURVE

CCW $\curvearrowleft_{C2}$ PRODUCES (-) JUMP IN M CURVE



- ③ SLOPE AND AREA RELATIONS



$$\Delta V = - \int_A^B w dx \quad \left\{ \begin{array}{l} \text{Change in } V \text{ over interval } AB \text{ is equal to} \\ \text{the area } [F/L][L] = [F] \text{ under the load curve.} \end{array} \right.$$

$$\frac{dV}{dx} = -w \quad \left\{ \begin{array}{l} \text{Slope of } V \text{ curve} = \text{neg. of load in interval.} \end{array} \right.$$

$$\Delta M = \int_A^B V dx$$

$$\frac{dM}{dx} = V$$

MATTERS OF STYLE

1. Find areas of each distributed load on beam, and areas of each interval on V curve. Write these on area and enclose w/ circle, e.g., (10), (15) etc.
2. Label all maxima, minima, + significant points.
3. Shade areas.
4. Label graphs w/ V & M and give units.

A useful formula for a parabolic area:

